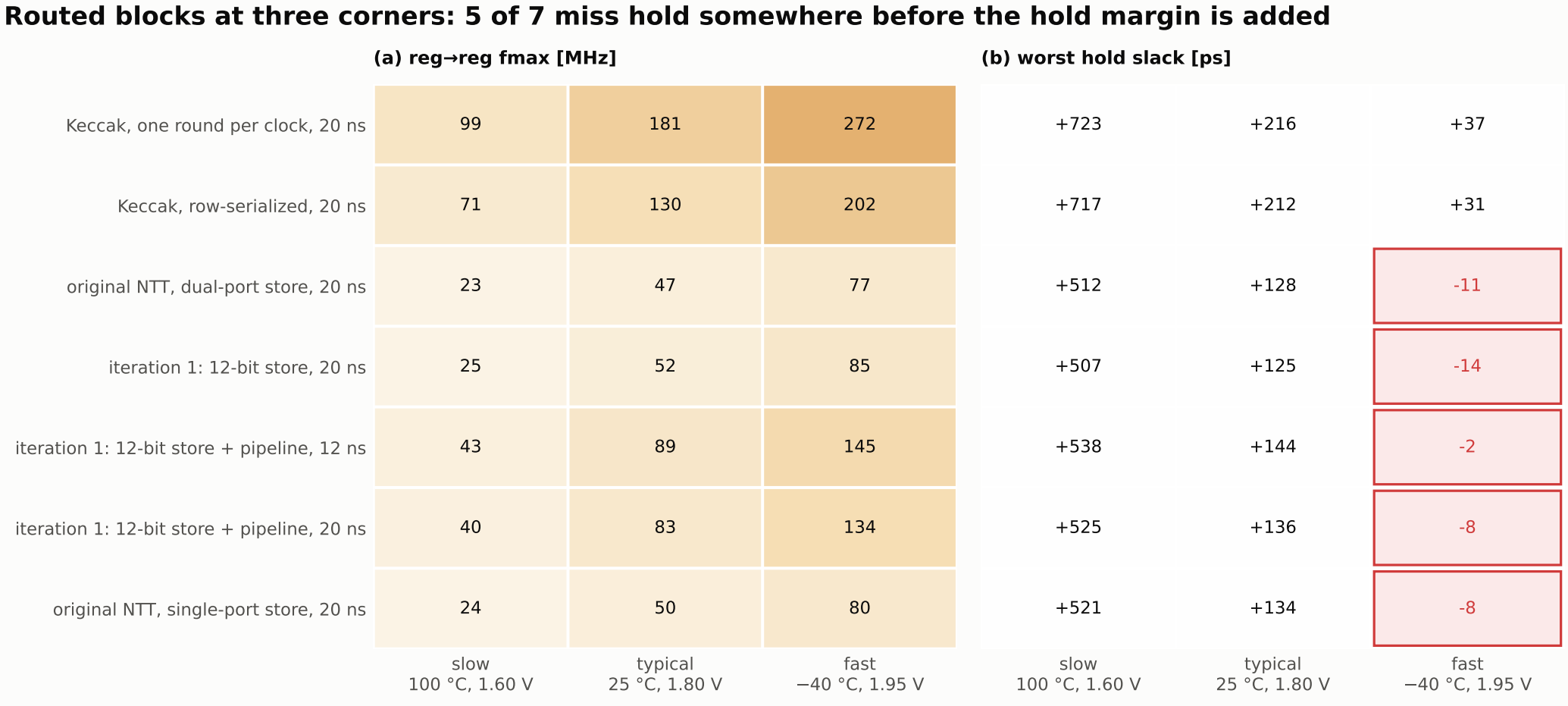
Routed blocks at three corners: 5 of 7 miss hold somewhere before the hold margin is added
(a) reg→reg fmax [MHz]
(b) worst hold slack [ps]
Keccak, one round per clock, 20 ns
99
181
272
+723
+216
+37
Keccak, row-serialized, 20 ns
71
130
202
+717
+212
+31
original NTT, dual-port store, 20 ns
23
47
77
+512
+128
-11
iteration 1: 12-bit store, 20 ns
25
52
85
+507
+125
-14
iteration 1: 12-bit store + pipeline, 12 ns
43
89
145
+538
+144
-2
iteration 1: 12-bit store + pipeline, 20 ns
40
83
134
+525
+136
-8
original NTT, single-port store, 20 ns
24
50
80
+521
+134
-8
slow
100 °C, 1.60 V
typical
25 °C, 1.80 V
fast
−40 °C, 1.95 V
slow
100 °C, 1.60 V
typical
25 °C, 1.80 V
fast
−40 °C, 1.95 V
Routed blocks at their own targets, OpenSTA with extracted parasitics; red: hold violated (before the hold margin)